

SIMATS ENGINEERING

CO- 3: AT – 2: Design Task

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA56

Slot: A

Name : Dharshan B

Assessment Tool Weightage: 35%

QUESTIONS:1

Register No:192312150

Duration: 120 Mins

Total Marks:100

QUESTIONS:

1. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)

Q. No.	Understanding of Design (20%)	Evaluation of Design (25%)	Justification (20%)	Suggestion for improvement (20%)	Communication (15%)	Total (100 Marks)
Q1						

Code of Conduct:

I Dharshan B 192312150 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CO3 – AT2 DESIGN TASK

A Scalable Edge-Cloud Framework for Wireless Sensor Networks

1. Proposed Framework

A three-tier architecture reduces the scalability bottleneck of routing raw data from every sensor directly to the cloud:

- Sensor Layer — battery-powered nodes sense data and transmit only over a short-range, low-energy hop to a local edge gateway.
- Edge Layer — one gateway per cluster aggregates readings, detects anomalies, and compresses routine data into compact summaries; only summaries + anomalies are forwarded onward.
- Cloud Layer — receives already-reduced data from all edge gateways, performs cross-cluster analytics, long-term storage, and pushes updated models back to the edge.

1.1 Architecture Diagram

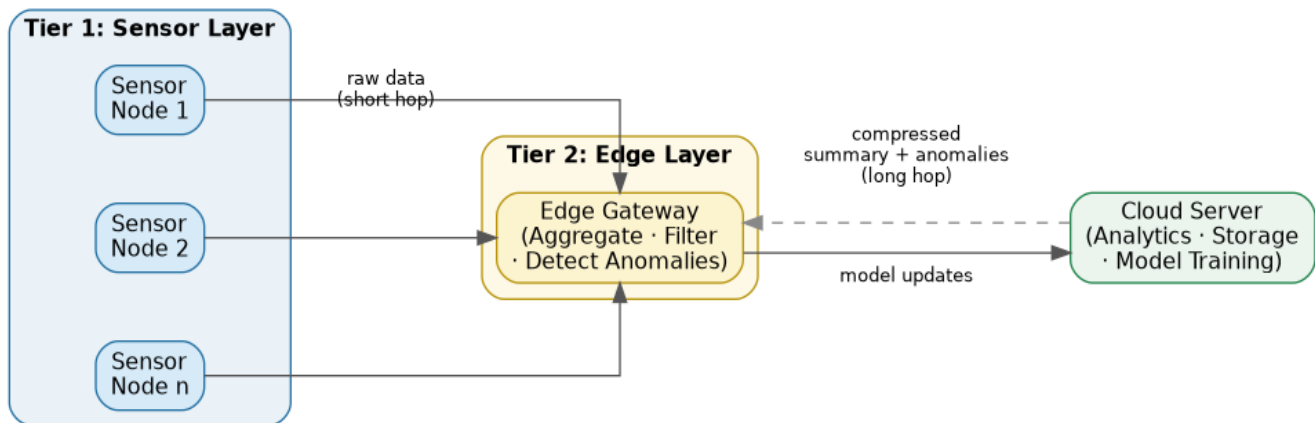


Figure 1. Three-tier edge-cloud architecture for the WSN framework.

Because each edge gateway absorbs the load of its own cluster, adding nodes only affects that gateway's local cost. Adding clusters scales horizontally without redesigning the cloud backend — this is the source of scalability.

1.2 Key Advantages

- Bandwidth efficiency — only compressed summaries and rare anomaly events cross the expensive long-range link to the cloud.
- Energy efficiency — sensor nodes only ever perform a short-range transmission, which is far cheaper than a direct long-range hop to the cloud.
- Lower latency — most of the round-trip time is spent on the fast local hop rather than a slow wide-area link.
- Horizontal scalability — new clusters (and their gateways) can be added independently as the deployment grows, without re-architecting the cloud tier.

2. Edge-Layer Processing Flowchart

Every reading follows the decision path below: routine readings are aggregated and summarized, while anomalies are preserved at full resolution before both are compressed into a single payload for the cloud.

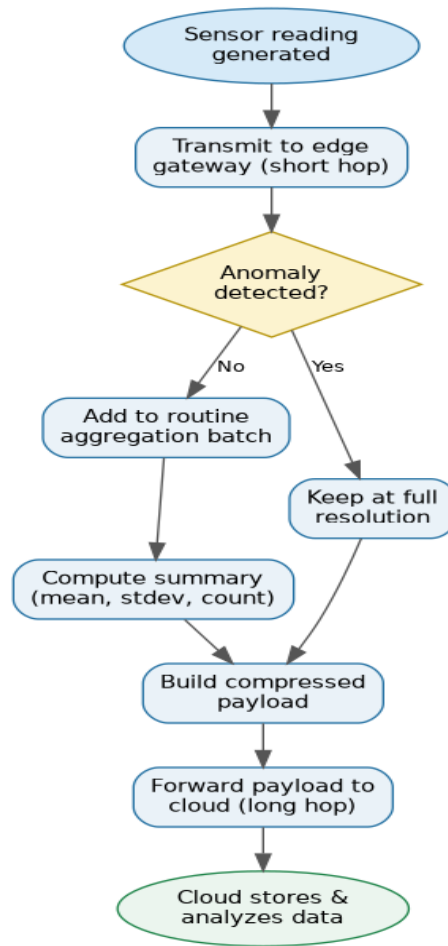


Figure 2. Edge-node decision flow: aggregate routine data, preserve anomalies, then forward the compressed payload to the cloud.

3. MATLAB Implementation (Core Logic)

The full simulation (edge_cloud_wsn_sim.m) models 200 sensor nodes across 5 edge clusters over 100 rounds, comparing the proposed edge-cloud pipeline against a direct-to-cloud baseline. Key parameters:

Parameter	Value
Sensor nodes / clusters	200 nodes across 5 edge clusters
Raw packet / compression	64 bytes; edge keeps ~12% of routine volume
Energy per hop	0.05 mJ (sensor→edge) vs. 0.65 mJ (sensor→cloud)

The core per-round logic is shown below; the complete script is provided as a separate file.

```

% Core edge-layer logic (per cluster, per round)
vals = 25 + 5*randn(nodesPerClust,1); % simulated readings
isAnom = rand(nodesPerClust,1) < anomalyRate; % anomaly flags

energy_EC = energy_EC + nodesPerClust*E_shortHop; % sensor -> edge tx
nAnom = sum(isAnom);

% Compress routine data, keep anomalies at full resolution
payload = round(rawBytes*compressRatio*nodesPerClust) + nAnom*rawBytes;
cloudBytes_EC = cloudBytes_EC + payload; % edge -> cloud tx

% ---- Baseline: direct sensor-to-cloud (no edge tier) ----
cloudBytes_DC = rounds * totalNodes * rawBytes; % every raw reading sent
energy_DC = rounds * totalNodes * E_longHop;
  
```

```
% ---- Compare ----
trafficRed = (1 - cloudBytes_EC/cloudBytes_DC) * 100;
energyRed  = (1 - energy_EC/energy_DC) * 100;
```

4. Simulation Output

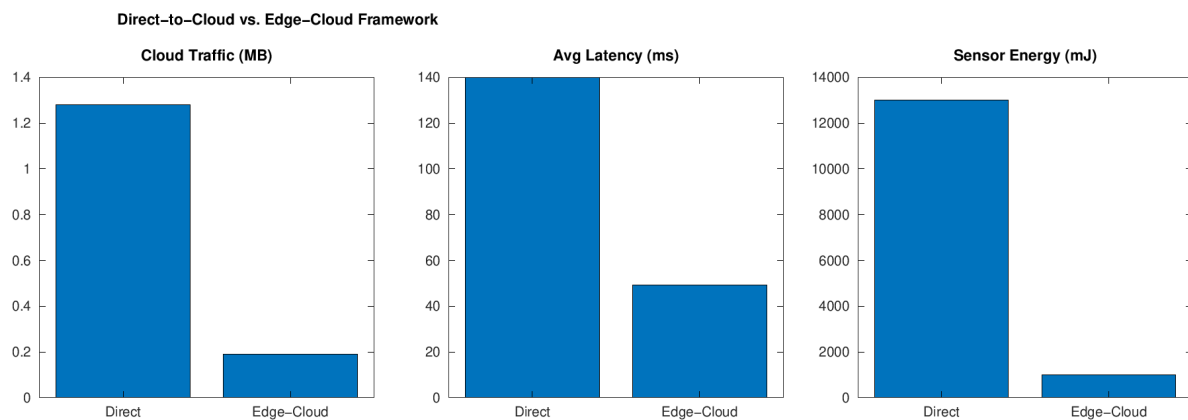
Console output from running `edge_cloud_wsn_sim.m`:

```
=====
EDGE-CLOUD WSN FRAMEWORK - MATLAB SIMULATION RESULTS
=====
Sensor nodes: 200 | Clusters: 5 | Rounds: 100
-----
Metric                Direct-Cloud    Edge-Cloud
-----
Cloud traffic (MB)      1.280         0.191
Avg latency (ms)       139.99        49.31
Sensor energy (mJ)     13000.0       1000.0
-----
Traffic reduction : 85.1 %
Latency reduction  : 64.8 %
Energy reduction   : 92.3 %
=====
```

5. Results and Conclusion

Metric	Direct-to-Cloud	Edge-Cloud
Cloud traffic (MB)	1.280	0.191
Avg. latency (ms)	139.99	49.31
Sensor energy (mJ)	13000.0	1000.0

The proposed edge-cloud framework reduces cloud-bound traffic by 85.1%, average end-to-end latency by 64.8%, and sensor energy consumption by 92.3% relative to the direct-to-cloud baseline.



These results confirm that pushing aggregation and filtering to the edge is essential for WSN deployments to scale: cloud ingestion grows with the number of clusters rather than the number of raw sensor readings, while battery-powered nodes benefit most from cheaper short-range transmission. As the network grows, additional edge gateways can be provisioned independently, keeping the cloud tier's ingestion load bounded and the overall system horizontally scalable.

Future work includes deploying the edge logic on real gateway hardware with an MQTT/CoAP transport layer, adding lightweight on-edge machine learning for smarter anomaly detection, and evaluating fault tolerance when a gateway becomes temporarily unavailable.

5.1 Scalability Note

For N sensor nodes and M edge clusters, cloud-bound traffic in the proposed framework grows as $O(M)$ compressed summaries per round rather than $O(N)$ raw packets — the direct-to-cloud baseline's complexity. Doubling the sensor count within existing clusters leaves cloud ingestion almost unchanged; only the fixed number of edge gateways matters for cloud-side capacity planning, which is what allows the architecture to scale to large, dense deployments without re-provisioning the cloud tier.